



National PG College, Lucknow – 226001
(An Autonomous College of University of Lucknow)

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Department of Zoology



Department of Zoology

Nomenclature of Syllabi

(As per NEP, 2020)

From the Session 2026-27



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Department of Zoology



CURRICULUM
&
PROGRAM STRUCTURE
B. Sc. Zoology

Purpose of the program:

Zoology as one of the subjects at undergraduate level should be studied in an integrated and cross- disciplinary manner with a comprehensive understanding of all living systems and their relationship with the ecosystem. Within the broad- range skill sets related to the discipline, it is required to impart and assess the quality of critical thinking, analytical and scientific reasoning, and problem-solving capacity.

Program Outcomes:

(As per Bloom's Taxonomy)

PO1 - Disciplinary knowledge

- Identify fundamental concepts, theory, principles, laws and definitions of disciplines.
- Explain scientific principles, theorems and laws and differentiate between them.
- Apply basic theories and principles to solve the problems in laboratory and real life problems.

PO2 – Communication Skills

- Express thoughts and ideas effectively in oral and writing.
- Demonstrate complex information in clear and concise manner to different groups.
- Ability to communicate with others using appropriate media confidently, share one's views and express herself /himself.

PO3 – Critical Thinking

- Apply informed actions after identifying the assumptions that frame our thinking and actions.
- Analyse our ideas and decisions (intellectual, organizational, and personal) from different perspectives.
- Evaluate the degree to which these assumptions are accurate and valid.



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PO4 – Problem Solving

- Explain problem statements, interpret requirements, and outline possible approaches.
- Identify and solve problems of relevance to society to meet the specified needs using the knowledge, skills and attitudes acquired from various discipline.
- Analyze complex problems into smaller parts, identify causes, and compare alternative solution strategies by using fundamental concepts, mathematical tools, and scientific methods to solve structured problems.

PO5 - Analytical Reasoning

- Identify the fundamental concepts, definitions, and principles relevant to scientific problems and find relationships among concepts with logical reasoning.
- Apply analytical methods and mathematical tools to interpret scientific information and figures.
- Break down complex problems into smaller, logical components for systematic study and compare alternative solutions using evidence and structured reasoning.

PO6 - Moral and Ethical Awareness/Reasoning

- Discuss ethical principles, codes of conduct and moral values relevant to science and society
- Explain the importance of honesty, integrity and respect for diversity in academic and professional settings.
- Demonstrate ethical practices in laboratory and research, and make use of technology to differentiate between ethical and unethical practices in real life applications

PO7 - Cooperation/Team work

- Collaborate to find solution of problems, roles and responsibilities in a team and explain the importance of co-operation and respect.
- Contribute actively in team work and function productively to achieve a common objective.
- Resolve and analyze team performance.

PO8 - Scientific Reasoning

- Discuss laws, principles, and theories relevant to natural and applied sciences.
- Explain natural phenomenon using scientific reasoning and demonstrate analogous relationship.
- Apply scientific methods to design experiments, test hypotheses, and apply concepts in laboratory or field settings and analyze experimental results to identify variables, patterns, errors, and logical inconsistencies.

PO9 - Information/Digital Literacy

- Develop the capability to use ICT in a variety of learning situations in academic and laboratory.



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- Demonstrate the potential to access, evaluate and use a variety of relevant information sources.
- Apply digital tools and simulation software for data analysis and compare results for credibility, relevance and accuracy.

PO10 - Life-long Learning

- Acquire the ability to engage in independent and lifelong learning in the broadest context socio-technological changes
- Exhibit the ability to stay competitive and employable by continuously updating skills and knowledge.
- Practice emerging trends and compare them with existing knowledge using online resources, journals and workshops.

Programme Specific Outcomes (PSOs):

- **PSO1-** Discuss the core concepts of animal diversity, morphology, physiology, ecology, evolution, parasitology, microbiology, cell biology, genetics, and biotechnology. Apply this knowledge to conduct laboratory experiments, dissections, ecological surveys and data analysis using modern biological tools and techniques.
- **PSO2-** Acquire and exhibit practical and vocational skills in areas like apiculture, sericulture, aquaculture, vermiculture, microbiology, bioinformatics, biostatistics, parasitology, biotechnology and wildlife conservation. These skills make them employable in various sectors including agriculture, fisheries, forestry, public health, environmental management and biodiversity conservation.
- **PSO3-** Develop understanding of the role of animals in ecosystems, conservation strategies, environmental awareness, and sustainable use of resources. Along with scientific ethics and digital literacy, students will cultivate the ability to engage in lifelong learning, adapt to socio-technological changes and work responsibly by applying zoological knowledge in addressing issues of biodiversity loss, climate change, pollution control, and public health challenges.



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B.Sc. (Zoology)

Year	Sem.	Code	Title of Paper	Theory/ Practical	Credits	Hours
Certificate Course in Animal Diversity						
I	I	ZOO101	Non-Chordata (Protozoa to Helminthes)	Theory	3	45
		ZOO102	Non-Chordata (Annelida to Echinodermata)	Theory	3	45
	II	ZOO201	Diversity and Biology of Chordata	Theory	4	60
		ZOO202	Practical	Practical	2	60
Diploma in Applied Zoology and Animal Conservation						
II	III	ZOO301	Environmental Biology and Wildlife	Theory	4	60
		ZOO302	Practical	Practical	2	60
	IV	ZOO401	Applied Zoology	Theory	3	45
		ZOO402	Microbiology, Bioinformatics, Biostatistics, Biological Tools	Theory	3	45
Degree in Bachelor of Science						
III	V	ZOO501	Animal Physiology and Biochemistry	Theory	4	60
		ZOO502	Evolutionary Biology and Biosystematics	Theory	4	60
		ZOO503	Practical	Practical	2	60
	VI	ZOO601	Cytogenetics and Molecular Biology	Theory	4	60
		ZOO602	Neurobiology, Physiological Constraints, Biotechnology, Toxicology	Theory	4	60
		ZOO603	Practical	Practical	2	60



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SEMESTER – I (Paper-I)

Paper Code	ZOO101	L	T	P	C
Paper Name	Non-Chordata (Protozoa to Helminthes)	3	0	0	0
Learning Objective	-To make the students aware of basic understanding of invertebrate animals. -Facilitate the students to deep dive into studies of animal diversity. -To develop basic understanding in students about evolutionary relationships among animal phyla.				
Course Outcomes	CO1: Describe the general characters, classification and diversity of lower invertebrates (Protozoa, Porifera, Coelenterata, Platyhelminthes, Nematelminthes). CO2: Explain the morphology, anatomy, physiology, reproduction and life cycles of representative types (<i>Paramecium</i> , <i>Sycon</i> , <i>Obelia</i> , <i>Fasciola</i> , <i>Ancylostoma</i>). CO3: Analyze structural and physiological adaptations, including polymorphism, canal systems, metagenesis, and parasitism in different phyla. CO4: Develop comparative understanding of organ systems (digestive, excretory, nervous, reproductive, respiratory) across lower invertebrates. CO5: Apply zoological knowledge to understand pathogenicity, economic importance, parasitic adaptations, disease transmission, treatment and prevention in Platyhelminthes. CO6: Demonstrate awareness of the ecological significance, biodiversity, and conservation importance of lower invertebrates in ecosystems.				

Programme: B.Sc. Zoology		Year: First	Semester: First
Subject: Zoology (Paper-I)			
Course Code: ZOO101		Course Title: Non-Chordata (Protozoa to Helminthes)	
Credits: 3	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 45			
Unit	Topics		No. of Lectures



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I	Phylum Protozoa – Characters, classification and types. Paramecium – Habit and habitat, structure, nutrition, digestion, excretion, respiration, osmoregulation, behavioral response, reproduction, binary fission, conjugation, autogamy, cytogamy, hemixis, endomixis.	11
II	Phylum Porifera - Characters, classification and types. Sycon – Habit and habitat, external structure, canal system, body wall, nutrition, digestion, excretion, respiration, reproduction and development, types of water canal system.	11
III	Phylum Coelenterata - Characters, classification and types. Obelia – Habit and habitat, morphology, polymorphism, development of medusa, physiology, reproduction, life cycle, alternation of generation and metagenesis.	11
IV	Phylum Platyhelminthes – Characters, classification and types. Fasciola – Habit and habitat, morphology, body wall, digestive system, excretory system, respiration, nervous system, reproduction, life history, types of larval stages, economic importance of <i>Fasciola</i> , parasitic adaptation of <i>Fasciola</i> . Role of lower invertebrates in Ecosystem Sustainability; Traditional Indian Knowledge (IKS) about Parasitic diseases	12
Suggested Readings: • Ruppert, EE, Fox R.S., Barnes R.D. (2004) Invertebrate Zoology, 7th Edition. Cengage Learning • Thomas Jeffrey Parker, William A. Haswell (2016). Parker & Haswell's A Textbook of Zoology Volume 1. WENTWORTH Press • Brusca (2016). Invertebrates. Sinauer • Pechenik Jan (2014). Biology of the invertebrates. McGraw Hill • Barnes R. S. K., Calow P. P., Olive P. J. W., Golding D. W., Spicer J. I. (2009). The Invertebrates: A Synthesis. Wiley Blackwell • Kotpal R.L. (2018) Modern Text Book of Zoology: Invertebrates. Rastogi Publications • Jordan E.L. and Verma P.S.: Invertebrate Zoology. S. Chand Publications.		



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CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	1	2	1	2	1	1	2	1	2	3	2	2
CO2	3	1	2	2	3	1	1	3	1	2	3	2	2
CO3	3	1	2	3	3	1	1	3	1	2	3	3	2
CO4	2	2	3	2	3	1	2	2	2	2	3	2	2
CO5	2	1	2	3	2	3	2	2	2	2	2	3	3
CO6	2	2	3	2	2	3	2	3	2	3	2	2	3

(3: High Correlation, 2: Moderate Correlation, 1: Low Correlation, 0: No Correlation)



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B. Sc. (Zoology)

SEMESTER – I (Paper-II)

Paper Code	ZOO102	L	T	P	C
Paper Name	Non-Chordata (Annelida to Echinodermata)	3	0	0	0
Learning Objective	-To make the students understand evolutionary relationship among lower and higher groups of animals. -To make the students learn about similarities and differences among different animal phyla. -To develop in the students the understanding of structural and functional diversity.				
Course Outcomes	CO1: Describe the general characters, classification and diversity of Annelida, Arthropoda, Mollusca and Echinodermata. CO2: Explain the morphology, anatomy, physiology and reproduction of <i>Hirudinaria</i> , <i>Palaemon</i> , <i>Pila</i> and <i>Asterias</i> . CO3: Analyze the structural adaptations, sense organs, larval stages, and economic importance of representative organisms. CO4: Develop comparative understanding of organ systems (digestive, excretory, respiratory, circulatory, nervous) across different phyla. CO5: Apply zoological knowledge to understand parasitism, vector biology, insect economic importance, pearl formation, and their role in human welfare. CO6: Demonstrate awareness of the ecological roles, biodiversity significance, and conservation aspects of higher non-chordates.				

Programme: B.Sc. Zoology		Year: First	Semester: First
Subject: Zoology (Paper-II)			
Course Code: ZOO102		Course Title: Non-Chordata (Annelida to Echinodermata)	
Credits: 3	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 45			
Unit	Topics		No. of Lectures



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I	Phylum Annelida – Characters, classification and types. Hirudinaria – Habit and Habitat, morphology, segmentation, receptors, sense organs, division of body, body apertures, body wall, locomotion, alimentary, digestive system, excretory system, respiratory system, nervous system, coelom and haemocoelomic system, reproduction, parasitic adaptations in <i>Hirudinaria</i> , economic importance of <i>Hirudinaria</i> .	11
II	Phylum Arthropoda – Characters, classification and types. Palaemon – Habit and habitat, morphology, exoskeleton, appendages, body wall, digestive system, excretory system, respiratory system, nervous system, sense organs or receptors, compound eyes, mosaic vision, apposition images and superposition images.	12
III	Phylum Mollusca – Characters, classification and types. Pila – Habit and Habitat, external features, body wall, mantle, digestive system, excretory system, respiratory system, mode of respiration, nervous system, blood vascular system, pearl formation in mollusca.	11
IV	Phylum Echinodermata – Characters, classification and types. Asterias - Habit and Habitat, external features, oral and aboral views, body wall, digestive system, excretory system, respiratory system, nervous system, water vascular system, larval stages in Echinodermata. Role of lower invertebrates in Ecosystem Sustainability, Gender roles in Invertebrate (Annelida to Echinodermata) based livelihoods.	11

Suggested Readings:

- Ruppert, EE, Fox R.S., Barnes R.D. (2004) Invertebrate Zoology, 7th Edition. Cengage Learning
- Thomas Jeffrey Parker, William A. Haswell (2016). Parker & Haswell's A Textbook of Zoology Volume 1. WENTWORTH Press
- Brusca (2016). Invertebrates. Sinauer
- Pechenik Jan (2014). Biology of the invertebrates. McGraw Hill
- Barnes R. S. K., Calow P. P., Olive P. J. W., Golding D. W., Spicer J. I. (2009). The Invertebrates: A Synthesis. Wiley Blackwell



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- Kotpal R.L. (2018) Modern Text Book of Zoology: Invertebrates. Rastogi Publications
- Jordan E.S. and Verma P.S. : Invertebrate Zoology. S. Chand Publications.

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	1	2	1	2	1	1	2	1	2	3	2	2
CO2	3	1	2	2	3	1	1	3	1	2	3	2	2
CO3	3	1	2	3	3	1	1	3	1	2	3	3	2
CO4	2	2	3	2	3	1	2	2	2	2	3	2	2
CO5	2	1	2	3	2	3	2	2	2	2	2	3	3
CO6	2	2	3	2	2	3	1	2	1	1	2	2	1

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SEMESTER – II (Paper-I)

Paper Code	ZOO201	L	T	P	C
Paper Name	Diversity and Biology of Chordata	4	0	0	0
Learning Objective	-To educate students about the comprehensive identification abilities of chordate diversity. -To understand anatomical, physiological and behavioral differences among different vertebrate phyla.				
Course Outcomes	CO1: Describe the systematic position, morphology, anatomy and affinities of Hemichordates, Urochordates, and Cephalochordates. CO2: Explain the digestive, respiratory, circulatory, excretory, nervous and reproductive systems of <i>Balanoglossus</i> , <i>Herdmania</i> and <i>Amphioxus</i> . CO3: Analyze the developmental biology and larval forms (Tornaria larva, retrogressive development in <i>Herdmania</i> , <i>Amphioxus</i> embryology). CO4: Compare the affinities, evolutionary relationships and chordate features across protochordates and vertebrates. CO5: Classify vertebrates (Pisces to Mammalia) up to order level with examples, highlighting adaptive features and diversity. CO6: Demonstrate awareness of the ecological, evolutionary and conservation significance of chordates and vertebrates.				

Programme: B.Sc. Zoology		Year: First	Semester: Second
Subject: Zoology (Paper-I)			
Course Code: ZOO201		Course Title: Diversity and Biology of Chordata	
Credits: 4	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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I	Hemichordata: <i>Balanoglossus</i> – External morphology, shape, size and colour, habit and habitat, division of body – proboscis, collar, trunk, body wall, coelom, endoskeleton, digestive system (alimentary canal, food and feeding mechanism, digestion), respiratory system (branchial pharynx, branchial sac, mechanism of respiration), blood vascular system, excretory system, nervous system, reproductive system, affinities of <i>Balanoglossus</i> , Development – Fertilization, embryonic or pre-larval development, larval development, Tornaria larva, metamorphosis.	15
II	Urochordata: <i>Herdmania</i> : External morphology, shape, size and colour, habit and habitat, division of body – body proper, foot, test or tunic, spicules, mantle / body wall, internal organs / viscera, coelom and atrium, digestive system, alimentary canal, food and feeding mechanism, digestion, respiratory system, blood vascular system, course of blood circulation, excretory system, nervous system, neural complex, sense organs, reproductive system, affinities of Urochordata, Development of <i>Herdmania</i> – Fertilization, embryonic or pre-larval development, larval development, retrogressive metamorphosis.	15
III	Cephalochordata: <i>Amphioxus</i> : External morphology, shape, size and colour, habit and habitat, division of body, body apertures, body wall – skin, musculature, peritoneum, coelom, skeleton, digestive system (alimentary canal, food and feeding mechanism, digestive glands, physiology of digestion), respiratory system, blood vascular system, excretory system (Protonephridia, hatschek's nephridium, brown funnel, renal papillae), nervous system, sense organ / receptors, reproductive system.	15
IV	Classification of different classes of vertebrates Pisces, Amphibia, Reptilia, Aves, Mammalia up to order with characters and examples.	15



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Suggested Readings:

- Young, J. Z. (2004). The Life of Vertebrates. III Edition. Oxford University Press.
- Kenneth V. Kardong (2015). Vertebrates: Comparative Anatomy, Function, Evolution. McGraw Hill
- Thomas Jeffrey Parker, William A. Haswell (2016) Parker & Haswell's A Textbook of Zoology Volume 2. WENTWORTH Press
- Eroschenko, Victor P. (2008). diFiore's Atlas of Histology with Functional correlations. XII Edition. Lippincott W. & Wilkins

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	1	2	2	2	1	1	2	1	2	3	2	2
CO2	3	1	2	2	3	1	1	3	1	2	3	2	2
CO3	3	1	2	3	3	1	1	3	1	2	3	3	2
CO4	2	2	3	2	3	1	2	2	2	2	3	2	2
CO5	3	1	2	3	2	2	2	2	2	2	3	3	3
CO6	2	2	3	2	2	3	2	3	2	3	2	2	3

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SEMESTER – II (Paper-II)

Paper Code	ZOO202	L	T	P	C
Paper Name	Practical	0	0	2	0
Learning Objective	- This programme aims to introduce students to animal diversity of invertebrates. The students will be taught about invertebrates using observational strategies, museum specimens and field reports. - Keeping in mind career perspectives in private and public labs, the lab courses have been substituted to ensure high quality training of students in laboratories.				
Course Outcomes	CO1: Perform major dissections (nervous system of <i>Palaemon</i>, <i>Pila</i>, <i>Unio</i>) and identify important structures with scientific accuracy. CO2: Carry out minor dissections (nerve ring of earthworm) and prepare permanent and temporary slides (gemmules, parapodium, spicules, setae, statocyst, hastate plate, radula) for microscopic study. CO3: Demonstrate skills in the identification and classification of specimens and slides of non-chordates across all major phyla. CO4: Apply practical knowledge and laboratory skills in handling dissection tools, preparing slides, recording observations and maintaining a class record. CO5: Develop analytical and presentation skills through spotting, project work, assignments, and viva-voce, enhancing scientific communication. CO6: Exhibit awareness of ethical issues, sustainability and animal welfare in biological experimentation, while promoting teamwork and lifelong learning.				

Programme: B.Sc. Zoology		Year: Second	Semester: Second
Subject: Zoology (Paper-II)			
Course Code: ZOO202		Course Title: Practical	
Credits: 2	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
	Topics		No. of Lectures



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	1. Major dissection: i. Central Nervous System of <i>Palaemon</i>. ii. Central Nervous System of <i>Pila</i>. - iii Nervous System of <i>Unio</i> 2. Minor dissection: i. Nerve ring of earthworm 3. Permanent stained preparation of: i. Gemmules of Sponges ii. Parapodium of <i>Neries</i> iii. Septal nephridia of Earthworm 4. Temporary / Glycerine preparation of: i. Spicules of sponges ii. Setae (in situ) of Earthworm iii. Statocyst of <i>Palaemon</i>. iv. Hastate plate of <i>Palaemon</i> v. Radula of <i>Pila</i> 5. Spotting – Study of specimens and slides of Non-chordata (Protozoa, Porifera, Coelenterata, Platyhelminthes, Nematelminthes, Annelida, Arthropoda, Mollusca, Echinodermata). 6. Project / Assignment 7. Class record 8. Viva-voce	60
Suggested Readings: • P. S. Verma: Practical Invertebrate Zoology • S. S. Lal: Practical Invertebrate Zoology		

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	1	2	2	2	2	1	2	1	2	3	2	2
CO2	3	1	2	2	3	2	1	3	1	2	3	2	2
CO3	3	1	2	3	3	1	1	3	1	2	3	3	2



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CO4	2	2	2	2	3	2	2	2	2	2	3	2	2
CO5	2	3	3	2	2	2	2	2	2	3	2	3	3
CO6	2	2	3	2	2	3	3	3	2	3	2	2	3

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SEMESTER – III (Paper-I)

Paper Code	ZOO301	L	T	P	C
Paper Name	Environmental Biology and Wildlife	4	0	0	0
Learning Objective	-Inclusion of ecology and environmental sciences will enrich students with our world which is crucial for human wellbeing and prosperity. -To facilitate the understanding about environment, its components and importance. -To make the students aware of global environmental issues, policies and practices. -Learn, identify and differentiate between different categories of wild animals and their role to us.				
Course Outcomes	CO1: Explain the concept, structure, and functions of ecosystems, including food chains, food webs, ecological pyramids, and ecological niches. CO2: Describe ecological succession and analyze different theories and patterns of succession with examples (xerosere, hydrosere, mesosere). CO3: Identify and evaluate types, causes, and effects of environmental pollution (air, water, soil, noise, radiation) and suggest control measures. CO4: Demonstrate awareness of global environmental issues such as greenhouse effect, ozone depletion, and acid rain, along with conservation and sanitation practices. CO5: Analyze the wildlife of India, with emphasis on endangered species, their causes of decline, IUCN categories, and the Red Data Book. CO6: Critically evaluate wildlife conservation efforts in India (sanctuaries, national parks, biosphere reserves, projects, in-situ & ex-situ strategies).				

Programme: B.Sc. Zoology	Year: Second	Semester: Third
Subject: Zoology (Paper-I)		
Course Code: ZOO301	Course Title: Environmental Biology and Wildlife	
Credits: 4	Major (Core)	Max. Marks: 30+70
Total No. of Lectures: 60		



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Unit	Topics	No. of Lectures
I	Ecology: Ecosystem – Concept, components, fundamental operations, energy, flow, food chain, food web, trophic levels, abiotic and biotic factors, ecological pyramids Ancient concepts of ecological balance (Panchabhuta, interdependence of species) Ecological Niche – Introduction, examples of ecological niche, ecotypes and ecotones. Ecological succession – Kinds of succession, patterns of succession – Xerosere, hydrosere, mesosere, Theories of succession – Monoclimax theory, Polyclimax theory, Climax pattern theory.	15
II	Pollution: Concept, sources, types of air, water, soil, noise and radiation pollution. Control of environmental pollution. Recent environmental issues and their ethical implications. The Green house effect, Ozone depletion, Acid rain. Environmental awareness including water and resource conservation and sanitation.	15
III	Wild life of India: Knowledge of endangered species (Reptiles, birds and mammals), causes of being endangered, economic importance. IUCN categories and Red Data Book, Indian environmental laws and International Conventions	15
IV	Conservation of Wild Life: Traditional conservation ethics and practices in IKS (Ahimsa, Jeeva-Daya, Sacred groves, Bishnoi practices), Important sanctuaries, National parks and Biosphere Reserves of India. Current status of different projects launched for the conservation of animal species, in-situ and ex-situ conservation of wild life. Role of UN Sustainable Development Goals (SDGs) in conservation of wildlife.	15
Suggested Readings: • Odum E.P. (2005) Fundamentals of Ecology. Cengage Learning India Private Limited		



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- Smith Thomas M., Smith Robert Leo (2014) Elements of Ecology. Pearson Education
- Krebs, Charles J. 2009. Ecology: the experimental analysis of distribution and abundance. Pearson.
- Sharma PD (2018). Fundamentals of Ecology. Rastogi Publications.
- Sharma PD (2018). Environmental Biology and Toxicology. Rastogi Publications.
- Pepper, I.L., Gerba, C.P. & Brusseau, M.L. 2006. Environmental & Pollution Science. Elsevier Academic Press.
- Gupta, K.R. 2006. Environmental Legislation in India. Atlantic Publishers and Distributors.

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	1	1	2	1	2	3	2	2
CO2	3	2	3	2	2	1	1	2	1	2	3	2	2
CO3	3	2	3	3	3	2	2	3	2	3	3	3	3
CO4	2	2	3	2	3	3	2	3	2	3	2	3	3
CO5	3	2	3	2	2	2	2	3	2	2	3	3	3
CO6	3	2	3	2	2	1	1	2	1	2	3	2	2

(3: High Correlation, 2: Moderate Correlation, 1: Low Correlation, 0: No Correlation)



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B. Sc. (Zoology)

SEMESTER – III (Paper-II)

Paper Code	ZOO302	L	T	P	C
Paper Name	Practical	0	0	2	0
Learning Objective	-To provide hands-on training to observe anatomical features of animals of economic importance. -To provide knowledge of vertebrates through slides and museum specimens to enhance the student's understanding.				
Course Outcomes	CO1: Perform major and minor dissections of <i>Scoliodon</i> , identifying cranial nerves, branchial arteries, and internal organs with accuracy. CO2: Prepare and analyze permanent and temporary stained preparations (e.g., placoid scales, ampullae of Lorenzini, muscle fibers) to study microscopic anatomy. CO3: Demonstrate the use of physiological apparatus such as kymograph and chromatography in experimental zoology. CO4: Identify, classify, and describe specimens and slides of chordates, including endangered species, supported by spotting and osteology studies. CO5: Develop practical scientific skills through project/assignment work, histological analysis, and record preparation. CO6: Exhibit communication, critical thinking, and ethical responsibility in viva-voce, conservation awareness, and applied Zoology practices.				

Programme: B.Sc. Zoology		Year: Second	Semester: Third
Subject: Zoology (Paper-II)			
Course Code: ZOO302		Course Title: Practical	
Credits: 2	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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	- Major dissection: Vth, VIIth, IXth and Xth Cranial Nerves of <i>Scoliodon</i> - Minor dissection: - Afferent branchial arteries of <i>Scoliodon</i> - Efferent branchial arteries of <i>Scoliodon</i> - Internal ear of <i>Scoliodon</i> - Permanent stained preparation of - Placoid scales (Unstained) - Ampullae of Lorenzini - Histology Stained preparation of striped / unstriped muscle fibres. - Comment upon physiological apparatus - Kymograph - Chromatography - Spotting – Study of specimens and slides (Protochordata, Cyclostomata, Pisces, Amphibia, Reptilia, Aves, Mammalia), Models and Pictures of Endangered Animals. - Osteology – Study of bones of Frog, <i>Varanus</i>, Fowl and Mammals - Project / Assignment - Class Record - Viva-voce	60
Suggested Readings: P. S. Verma: Practical Invertebrate Zoology S. S. Lal: Practical Invertebrate Zoology		

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	1	2	2	1	2	3	3	2
CO2	3	2	3	2	2	1	1	2	1	2	3	3	2
CO3	3	2	3	3	2	2	1	2	2	3	3	3	2
CO4	3	2	3	2	2	2	2	3	2	2	3	3	3
CO5	2	2	3	3	3	2	2	3	2	3	3	3	3
CO6	2	3	2	2	3	3	3	3	3	3	3	3	3

(3: High Correlation, 2: Moderate Correlation, 1: Low Correlation, 0: No Correlation)



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B. Sc. (Zoology)

SEMESTER – IV (Paper-I)

Paper Code	ZOO401	L	T	P	C
Paper Name	Applied Zoology	3	0	0	0
Learning Objective	-To make students understand the economic loss caused by parasites, vectors and pests. -To make students aware about different career prospective in possible associated and applicable fields. -To make students understand the importance of wild life and its conservation.				
Course Outcomes	CO1: Describe the morphology, life cycle, pathogenicity, symptoms, and control measures of parasites of humans and domestic animals (<i>Dipyllobothrium</i> , <i>Hymenolepis</i> , <i>Dracunculus</i> , <i>Wuchereria</i> , <i>Trypanosoma</i> , <i>Giardia</i> , <i>Paragonimus</i>). CO2: Analyze the role of vectors and pests (mosquito, housefly, bed bug, louse, termite, cockroach, cloth moth, wax moth, Gundhi bug, sugarcane leaf hopper) and propose appropriate management and control strategies. CO3: Demonstrate knowledge of apiculture, sericulture, and lac culture, highlighting techniques, economic importance, and challenges in production. CO4: Apply the principles of aquaculture, poultry farming, and vermiculture for sustainable development and entrepreneurship. CO5: Develop analytical, problem-solving, and technical skills through projects, case studies, and practical applications of applied zoology. CO6: Exhibit ethical values, teamwork, and lifelong learning skills in handling parasitic diseases, pest control, and economic Zoology practices.				

Programme: B.Sc.	Year: Second	Semester: Fourth
Subject: Zoology (Paper-I)		
Course Code: ZOO401	Course Title: Applied Zoology	
Credits: 3	Major (Core)	Max. Marks: 30+70
Total No. of Lectures: 45		



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Unit	Topics	No. of Lectures
I	Parasitology: Structure, life cycle, pathogenicity including diseases, causes, symptoms and control of the following parasites of domestic animals and humans : <i>Diphyllbothrium</i> , <i>Hymenolepis</i> , <i>Dracunculus</i> , <i>Wuchereria</i> , <i>Trypanosoma</i> , <i>Giardia</i> , <i>Paragonimus</i>	11
II	Vectors and pests: Vectors – Mosquito, housefly, bed bug, louse and their control. Pest –Features, life cycle, nature of damage and control of termite, cloth moth, wax moth, Gundhi bug, sugarcane leaf hopper. Role of Sustainable Development Goals (SDGs) in ethically preventing diseases from pest, vector and parasites.	12
III	Industrial Zoology-I: Apiculture, Sericulture, Lac culture. Ancient Indian References, practices and sustainable use of beneficial insects (Indian Knowledge System)	11
IV	Industrial Zoology-II: Aquaculture, Poultry, Vermiculture. Ancient Indian References, practices and sustainable use of Matsya Palan, Pakshi Palan and Krimi Palan (Indian Knowledge System).	11
Suggested Readings: - Nigam H C (2014) Emerging Trends in Biology & Economic Zoology. Vishal Publishing Co. - Shukla GS & Upadhyay VB (2017) Economic Zoology Rastogi Publications - Srivastava KP and Dhaliwal GS. Textbook of Applied Entomology Volume 1 & 2. Kalyani Publishers. - Caughley, G., and Sinclair, A.R.E. (1994). Wildlife Ecology and Management. Blackwell Science. - Simpson: Principles of Animal Taxonomy (1962, Oxford). - Mayer & Ashlock: Principles of Systematic Zoology (2nd Edition, McGraw Hill). - Kapoor: Theory and Practicals of Animal Taxonomy (1988, Oxford & IBH).		



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- Zar JH (2010) Biostatistical Analysis. 5th Edition. Pearson.

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	3	2	3	2	2	3	3	2
CO2	3	2	3	3	2	3	2	3	2	2	3	3	3
CO3	3	2	2	3	2	2	2	2	2	3	3	3	3
CO4	3	2	2	3	3	2	2	2	2	3	3	3	3
CO5	2	2	3	3	3	2	2	3	3	3	3	3	3
CO6	2	3	2	2	3	3	3	3	3	3	3	3	3

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SEMESTER – IV (Paper-II)

Paper Code	ZOO402	L	T	P	C
Paper Name	Microbiology, Bioinformatics, Biostatistics, Biological Tools	3	0	0	0
Learning Objective	-Make students aware of various industrial biological fields to which animals are associated. -To know about basic scientific tools to manage the scientific data. -Make students learn about working of biological tools essential in research.-To make students understand the importance of wild life and its conservation.				
Course Outcomes	CO1: Explain the distribution, morphology, physiology, and infections caused by microorganisms in air, soil, water, milk, fruits, and vegetables, along with the role of antibiotics. CO2: Demonstrate knowledge of bioinformatics resources for scientific data storage, retrieval, and citation. CO3: Apply biostatistical methods (sampling, central tendency, dispersion, chi-square, correlation, regression) for biological data analysis. CO4: Illustrate the principle and applications of biological tools such as electrophoresis, pH meter, and different types of microscopy (light, SEM, TEM). CO5: Develop analytical reasoning and problem-solving skills by integrating microbiology, bioinformatics, and statistics in experimental and applied contexts. CO6: Exhibit ethical awareness, teamwork, and lifelong learning in handling microbes, using digital tools, and applying statistical approaches in biology.				

Programme: B.Sc. Zoology	Year: Second	Semester: Fourth
Subject: Zoology (Paper-II)		
Course Code: ZOO402	Course Title: Microbiology, Bioinformatics, Biostatistics, Biological Tools	
Credits: 3	Major (Core)	Max. Marks: 30+70
Total No. of Lectures: 45		



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Unit	Topics	No. of Lectures
I	Microbiology: Ancient knowledge of microorganisms in India (IKS), Distribution of microorganism in air, soil, water, milk, fruits and vegetables. Morphology, physiology and infection (outline) of bacterial and viral diseases, common antibiotic types, classification and mode of action.	11
II	Bioinformatics: Basic Scientific data resources (encyclopedia, biological abstracts, reviews, zoological records, current contents etc.). Storage, retrieval and citation of scientific data. Ethics in bioinformatics	11
III	Biostatistics: Sampling, measure of central tendency (Mean, median and mode) and dispersion (Variance, standard deviation and standard error), Test of significance of mean, the Chi-square test, correlation and regression.	12
IV	Biological Tools: Principle and use of common physiological apparatus in electrophoresis, pH meter, Microscopy – Light, Scanning and transmission electron microscopes.	11
Suggested Readings: • Microbiology: An Introduction: Tortora GJ, Funke BR, Case CL. • Prescott's Microbiology: Willey J, Sherwood L, Woolverton CJ. • Bioinformatics: Sequence and Genome Analysis by David W. Mount. • Introduction to Bioinformatics: Lesk AM. • Introduction to Biostatistics by Dr. Pranab Kr. Banarjee • Biostatistics: Triola MM and Triola MF. • Bioinstrumentation by L. Veerakumari • Bioinstrumentation: Webster JG.		



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CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	3	1	3	2	2	3	3	2
CO2	3	2	2	2	2	2	1	2	3	3	3	3	3
CO3	3	2	3	3	3	2	2	3	2	3	3	3	3
CO4	3	2	2	3	3	2	2	3	2	3	3	3	3
CO5	3	2	3	3	3	2	2	3	2	3	3	3	3
CO6	2	3	2	2	3	3	3	3	3	3	3	3	3

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SEMESTER – V (Paper-I)

Paper Code	ZOO501	L	T	P	C
Paper Name	Animal Physiology and Biochemistry	4	0	0	0
Learning Objective	-To make the student aware of functioning of the body and its various organs. -To develop in the student an understanding of the various homeostatic systems of the body. -To develop in the student an understanding of regulation of function in the body -To know the basic need of nutrients for growth and development of the body.				
Course Outcomes	CO1: Explain the mechanisms of digestion, absorption, and respiration, including structural and functional aspects of organs involved. CO2: Describe the morphology and physiology of the circulatory system and excretory system, emphasizing regulation and homeostasis. CO3: Interpret the role of endocrine glands, their secretions, and their impact on growth, development, and metabolism. CO4: Classify and analyze the biochemical nature and role of carbohydrates, lipids, proteins, and enzymes in metabolism. CO5: Apply knowledge of physiology and biochemistry to understand disease mechanisms, health management, and experimental studies. CO6: Demonstrate ethical responsibility, analytical reasoning, teamwork, and lifelong learning in applying physiology and biochemistry concepts.				

Programme: B.Sc. Zoology		Year: Third	Semester: Fourth
Subject: Zoology (Paper-I)			
Course Code: ZOO501		Course Title: Animal Physiology and Biochemistry	
Credits: 4	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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I	Digestion and absorption – Definition, types of digestion – intracellular digestion, extracellular digestion; Structure of alimentary canal and associated digestive glands (liver, pancreas, gall bladder); Digestion in mouth, digestion in stomach, digestion in intestine, defecation; Absorption of carbohydrates, proteins, fats. Respiration – Introduction, external respiration, breathing, process of gaseous exchange in the air and blood during respiration, condition of gases in the blood, respiratory exchange in tissues, transportation of oxygen and carbon dioxide, carbon dioxide dissociation curve, tissue respiration, regulation of respiration, chloride shift / hamburger phenomenon.	12
II	Circulatory system – Introduction, structure of mammalian heart. Blood vascular system, arterial and venous system, heartbeat, functioning of heart, cardiac cycle, regulation of heart beat, heart sounds – lupp, dupp, pulse, blood pressure. Excretion – Introduction, structure of kidney, functions of kidney – Urine formation, glomerular filtration, selective reabsorption, tubular secretion, counter current mechanism, regulation of kidney function, urea formation – ornithine cycle.	12
III	Endocrine glands – Introduction, nature of endocrine glands, pituitary gland, thyroid gland, parathyroid gland, adrenal gland, thymus gland, pineal body.	10
IV	Biochemistry – General Chemistry and classification for carbohydrates, lipids and proteins. Enzymes – Introduction, exoenzyme, endoenzyme, classification, nomenclature, chemical nature of enzyme, mechanism of enzyme action, activation of enzyme, isoenzyme, coenzyme, holoenzyme, apoenzyme, cofactor, factor influencing enzyme action. Enzyme inhibitors- Reversible and irreversible, competitive and noncompetitive.	11



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Suggested Readings:

- Guyton, A.C. & Hall, J.E. (2006). Textbook of Medical Physiology. XI Edition. Hercourt Asia PTE Ltd. /W.B. Saunders Company.
- Tortora, G.J. & Grabowski, S. (2006). Principles of Anatomy & Physiology. XI Edition John Wiley & sons
- Christopher D. Moyes, Patricia M. Schulte 2016 Principles of Animal Physiology. 3rd Edition, Pearson Education,
- Ganong: Review of Medical Physiology, Lang Medical Publ.
- Boyer: Concepts in Biochemistry (3rd ed. 2006, Brooks/Cole)

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	3	1	3	2	2	3	3	2
CO2	3	2	2	3	2	2	2	3	2	3	3	3	2
CO3	3	2	2	3	2	2	2	3	2	3	3	3	3
CO4	3	2	3	3	3	2	2	3	3	3	3	3	3
CO5	3	2	3	3	3	2	2	3	3	3	3	3	3
CO6	2	3	2	2	3	3	3	3	3	3	3	3	3

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SEMESTER – V (Paper-II)

Paper Code	ZOO0502	L	T	P	C
Paper Name	Evolutionary Biology and Biosystematics	4	0	0	0
Learning Objective	-To make students know about the origin of life on earth and its possible extraterrestrial existence. -To enhance the knowledge about various sources of variations and their role in evolution. -Understanding of animal taxonomy and systematic and their application -Enhance the knowledge about molecular basis of animal taxonomy.				
Course Outcomes	CO1: Explain the historical background and theories of origin of life, including biochemical and extraterrestrial concepts. CO2: Demonstrate understanding of evidences of organic evolution from anatomy, embryology, physiology, cytology, genetics, and biogeography. CO3: Compare and evaluate theories of evolution (Lamarckism, Darwinism, Modern Synthetic Theory). CO4: Understand the principles of taxonomy and biosystematics, including ICZN rules and taxonomic operations. CO5: Apply knowledge of biological classification, species concepts, and mechanisms of speciation in understanding biodiversity. CO6: Develop critical reasoning, ethical responsibility, research aptitude, and lifelong learning in evolutionary biology and taxonomy.				

Programme: B.Sc. Zoology		Year: Third	Semester: Fifth
Subject: Zoology (Paper-II)			
Course Code: ZOO502		Course Title: Evolutionary Biology and Biosystematics	
Credits: 4	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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I	Evolutionary Biology: Historical background of evolution, Historical contributions of Zoologists (Indian Knowledge System), old and new theories of origin of life; Biochemical origin of life, extra-terrestrial life.	15
II	Evidences of Organic Evolution: Evidences from classification, evidences from comparative anatomy, evidences from embryology, evidences from vestigial organs, evidences from physiology and Biochemistry, evidences from enzymes and hormones, evidences from cytology, evidences from connecting links, evidences from zoogeography, evidences from genetics. Theories of Evolution: Lamarckism, Darwinism and Modern synthetic theory.	15
III	Introduction to taxonomy and biosystematics: Indigenous classification of animals and Folk taxonomy (Indian Knowledge System), Operations in taxonomy, Importance of taxonomy and biosystematics, International Codes of Zoological Nomenclature (ICZN), Aims and tasks of a taxonomist, Animal diversity: Alpha, Beta and Gamma.	15
IV	Biological Classification: Kinds of Zoological classification, Taxonomic (Linnean) hierarchy, Concept of species, Mechanism of speciation, Theories of biological classification, Kinds and components of classification.	15
Suggested Readings: • Futuyma, Douglas J. and Kirkpatrick Mark. Evolution (4th Edition) Sinauer • Veer Bala Rastogi (2017) Organic Evolution. Med Tech • Hall B.K. and Hallgrímsson B. (2008). Strickberger's Evolution. IV Edition. Jones and Bartlett Publishers Inc. • Huxley Julian. Evolution: The Modern Synthesis. Harper and Brothers. • Kapoor V.C. 1991. Theory and Practice of Animal Taxonomy. Oxford and IBH Publishing.		



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CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	2	1	3	2	2	3	2	2
CO2	3	2	3	3	2	2	1	3	2	3	3	3	2
CO3	3	2	3	3	2	2	2	3	2	3	3	3	3
CO4	3	2	2	3	3	2	2	3	3	3	3	3	3
CO5	3	2	3	3	3	2	2	3	3	3	3	3	3
CO6	2	3	3	2	3	3	3	3	3	3	3	3	3

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SEMESTER – V (Paper-III)

Paper Code	ZOO503	L	T	P	C
Paper Name	Practical	0	0	2	0
Learning Objective	-To provide hands-on training to observe anatomical features of animals of economic importance. -To provide knowledge of vertebrates through slides and museum specimens to enhance the student's understanding.				
Course Outcomes	CO1: Demonstrate dissection skills in cockroach CNS, salivary glands, mouthparts and explain their functional anatomy. CO2: Perform immunological tests such as detection of blood groups and interpret results. CO3: Carry out physiological experiments like haemin crystal preparation, blood film staining, and qualitative urine analysis. CO4: Prepare permanent and temporary mounts (arthropod mouthparts, parasitic larval stages, fish scales) and analyze their structures. CO5: Operate and explain the principle of physiological apparatus (haemocytometer, haemoglobinometer, pH meter). CO6: Develop practical research aptitude, teamwork, record-keeping, and critical analysis through assignments, projects, spotting, and viva.				

Programme: B.Sc. Zoology		Year: Third	Semester: Fifth
Subject: Zoology (Paper-III)			
Course Code: ZOO503		Course Title: Practical	
Credits: 2	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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	1. Major Dissection: Central nervous System of Cockroach 2. Minor Dissection • Salivary Gland complex of Cockroach • Mouth Parts of Cockroach 3. Immunology • Detection of blood group. 4. Physiology • Preparation of Haemin crystals. • Preparation of vertebrate blood film stained with Leishman's stain. • Detection of sugar / albumin / ketone bodies in an artificial sample of urine. 5. Permanent preparation • <i>Cimex</i> • <i>Pediculus</i> • <i>Haematopinus</i> • Scales of Fish 6. Temporary Preparation • Mouth parts of mosquito • Mouth parts of housefly • Redia Larva • Cercaria Larva 7. Comment upon physiological apparatus: • Haemocytometer • Haemoglobinometer • pH meter 8. Spotting- Study of specimens and slides of Protozoan parasites, Helminths, Arthropods, nematodes and edible fishes. Prepared slides of Histology and Embryology. 9. Assignment / Project 10. Class Record 11. Viva-voce	60
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Suggested Readings:

- P. S. Verma: Practical Invertebrate Zoology
- S. S. Lal : Practical Invertebrate Zoology

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	3	2	2	2	3	2	2	3	3	2
CO2	3	2	2	2	3	2	2	3	2	3	3	3	2
CO3	3	2	3	3	3	2	2	3	2	3	3	3	2
CO4	3	2	2	3	2	2	2	3	2	3	3	3	2
CO5	3	2	3	3	3	2	2	3	3	3	3	3	3
CO6	2	3	3	2	3	3	3	3	3	3	3	3	3

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SEMESTER – VI (Paper-I)

Paper Code	ZOO601	L	T	P	C
Paper Name	Cytogenetics and Molecular Biology	4	0	0	0
Learning Objective	-To understand the structure and functions of cell and its organelles. -To understand the structure and arrangement of genes and their interaction. -To understand the structure and types of DNA and its organization. -Enhance the knowledge about the life processes at cellular and molecular levels.				
Course Outcomes	CO1: Describe the structure, types, and functions of cell organelles (plasma membrane, nucleus, mitochondria, ER, Golgi, ribosome, lysosome). CO2: Explain chromosome structure, types, chemical composition, and special chromosomes (lampbrush, polytene) along with DNA/RNA structure. CO3: Apply Mendel's laws, linkage, and crossing over concepts to predict patterns of inheritance in different organisms. CO4: Analyze mechanisms of sex determination (chromosomal, environmental, hormonal, genic balance, cytoplasmic) and hereditary disorders in humans. CO5: Illustrate molecular processes including replication, transcription, translation, genetic code, RNA processing, and gene regulation. CO6: Develop critical thinking, problem-solving, and scientific reasoning skills in cell biology and genetics, relevant for research, healthcare, and biotechnology.				

Programme: B.Sc. Zoology		Year: Third	Semester: Sixth
Subject: Zoology (Paper-I)			
Course Code: ZOO601		Course Title: Cytogenetics and Molecular Biology	
Credits: 4	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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I	Cell Biology: Structure, types and function of cell and cell organelles (Plasma membrane, Mitochondria, Nucleus, Endoplasmic reticulum, Golgi Complex, ribosome and lysosomes)	15
II	Cell Biology: Chromosomes – Types of chromosomes, structure of chromosomes, chemical composition of chromosomes, functions of chromosomes, special type of chromosomes – Lampbrush Chromosomes, Polytene chromosomes. Watson and Crick model of DNA, differences between DNA and RNA.	15
III	Mendel's law of inheritance – Monohybrid cross, Dihybrid cross, Test Cross, Back cross, Incomplete dominance. Linkage – Introduction, coupling and Repulsion Hypothesis, examples of linkage, types of linkage, arrangement of linked genes, Chromosome theory of linkage, significance of Linkage. Crossing Over – Introduction, examples of crossing over, mechanism of crossing over, kinds of crossing over, percentage of crossing over, factors affecting crossing over, utility of crossing over. Sex Determination – genetically controlled sex determination, environmental role in sex determination, hormonal role in sex determination, chromosomal theory of sex determination, XX-Xy type / lygaeus type, XX-XO type / Protenor type. Quantitative theory of sex determination, genic balance theory, cytoplasmic sex determination, barr body / Lyons hypothesis. Hereditary diseases of men	15
IV	Nucleic Acids: structure, replication, central dogma, genetic code Protein synthesis (Transcription, Translation), RNA processing, Gene regulation	15



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Suggested Readings:

- Karp, G. (2010). Cell and Molecular Biology: Concepts and Experiments. VI Edition. John Wiley and Sons. Inc.
- De Robertis, E.D.P. and De Robertis, E.M.F. (2006). Cell and Molecular Biology. VIII Edition. Lippincott Williams and Wilkins, Philadelphia.
- Cooper, G.M. and Hausman, R.E. (2009). The Cell: A Molecular Approach. V Edition. ASM Press and Sunderland, Washington, D.C.; Sinauer Associates, MA.
- Gardner, E.J., Simmons, M.J., Snustad, D.P. (2008). Principles of Genetics. VIII Edition. Wiley India.

CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	2	1	3	2	2	3	2	2
CO2	3	2	2	2	2	2	1	3	2	2	3	2	2
CO3	3	2	3	3	3	2	2	3	2	3	3	3	2
CO4	3	2	3	3	3	2	2	3	2	3	3	3	3
CO5	3	2	3	3	3	2	2	3	3	3	3	3	3
CO6	2	3	3	3	3	3	3	3	3	3	3	3	3

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Department of Zoology



B. Sc. (Zoology)

SEMESTER – VI (Paper-II)

Paper Code	ZOO602	L	T	P	C
Paper Name	Neurobiology, Physiological Constraints, Biotechnology, Toxicology	4	0	0	0
Learning Objective	-To enhance the knowledge about neuronal conduction. -To get aware of psychophysiological changes in body during unfavorable conditions. -To develop an understanding about genetic engineering and its importance. -To make students understand about toxicants and their effects.				
Course Outcomes	CO1: Explain the structure and functions of neurons, glial cells, synapses, and mechanisms of nerve impulse transmission. CO2: Analyze the physiological effects of stress and weightlessness on different body systems. CO3: Demonstrate an understanding of genetic engineering, recombinant DNA technology, and applications of biotechnology in agriculture, medicine, food, and energy production. CO4: Illustrate the role of microbes in biotechnology (insulin production, fermentation, pharmaceuticals, food processing). CO5: Assess the toxicological principles including routes of exposure, dose-response relationships, and toxic effects. CO6: Develop critical thinking, ethical reasoning, and lifelong learning skills for applying neurobiology, biotechnology, and toxicology in research, health, and industry.				

Programme: B.Sc. Zoology		Year: Third	Semester: Sixth
Subject: Zoology (Paper-II)			
Course Code: ZOO602		Course Title: Neurobiology, Physiological Constraints, Biotechnology, Toxicology	
Credits: 4	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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I	Neurobiology: Neurons and glia and their types, synapse, propagation and conduction of nerve impulse (membrane and action potentials), synaptic potential, synaptic integration, synaptic transmission (electrical and chemical transmission).	15
II	Physiological constraints: Kinds of stress and their influence, physiological mechanism of stress. Neuroendocrine regulation of stress. Gender differences in stress response (HPA axis variations) Effect of weightlessness (In space environment) on neurosensory, cardiovascular and respiratory systems.	15
III	Biotechnology: Concept of Genetic engineering and recombinant DNA technology and its application in agriculture, medical areas and energy production. Biotechnology of food processing, Pharmaceuticals, uses of microbes in insulin production and fermentation. Ethics in Genetic engineering and Biotechnology	15
IV	Toxicology: Exposure of toxicants, routes of exposure, duration and frequency of exposure, dose response relationship, categories of toxic effects, local and systematic effects, immediate and delayed effects, reversible and irreversible effects, Traditional detoxification methods (IKS)	15
Suggested Readings: - Principles of Neural Science: Kandel ER, Schwartz J, Jessell T. - Fundamental Neuroscience: Squire LR, Bloom FE, Splitzer NC - Biotechnology: RC Dubey. S. Chand Publications - Genetic Engineering: TA Brown. - Stress Physiology in Animals: Balm PHM. Sheffield Academic. - Handbook of Toxicology, M.J. Derelanko & C.S. Auletta, 3rd Ed. CRC Press - Concepts of Toxicology. Omkar		



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CO-PO MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	2	1	3	2	2	3	2	2
CO2	3	2	3	3	2	2	2	3	2	2	3	2	2
CO3	3	2	3	3	3	2	2	3	3	3	3	3	3
CO4	3	2	2	3	3	2	2	3	3	3	3	3	3
CO5	3	2	3	3	3	3	2	3	2	3	3	3	3
CO6	2	3	3	3	3	3	3	3	3	3	3	3	3

(3: High Correlation, 2: Moderate Correlation, 1: Low Correlation, 0: No Correlation)



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B. Sc. (Zoology)

SEMESTER – VI (Paper-III)

Paper Code	ZOO603	L	T	P	C
Paper Name	Practical	0	0	2	0
Learning Objective	-To provide hands-on training to observe anatomical features of animals of economic importance. -To provide knowledge of vertebrates through slides and museum specimens to enhance the student's understanding.				
Course Outcomes	CO1: Demonstrate practical skills in major and minor dissections of vertebrates and invertebrates for understanding anatomical structures. CO2: Perform microbiological techniques such as culture and Gram staining of bacteria with accuracy and precision. CO3: Prepare and analyze cytological and histological slides including nerve cells, epithelial cells, and giant chromosomes. CO4: Apply molecular biology techniques such as DNA isolation and electrophoresis for genetic studies. CO5: Identify, analyze, and interpret specimens, life cycles, adaptations, and the impact of pollution/toxicology. CO6: Develop competence in handling biological tools, teamwork, scientific reasoning, record keeping, and project-based learning.				

Programme: B.Sc. Zoology		Year: Third	Semester: Sixth
Subject: Zoology (Paper-III)			
Course Code: ZOO603		Course Title: Practical	
Credits: 2	Major (Core)		Max. Marks: 30+70
Total No. of Lectures: 60			
Unit	Topics		No. of Lectures



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	1. Dissection – a. Take out Afferent branchial arteries, efferent branchial arteries, cranial nerves (Vth, VIIth, IXth and Xth and their branches) of <i>Wallago</i>. b. Central nervous system of <i>Periplaneta americana</i> (Cockroach). 2. Minor dissection – a. Salivary gland complex of Cockroach b. Webberian ossicle of <i>Wallago</i>. 3. Culture and gram staining of Bacteria. 4. Preparation of: a. Slide of nerve cells from given samples. b. Cheek epithelial cells with Methylene Blue. 5. Cytological preparation- Make a slide of salivary gland for giant chromosomes of <i>Chironomous</i> / <i>Drosophila</i> larva. 6. Molecular Biology – A. Isolation of genomic DNA from Animal cell. B. Electrophoresis of DNA samples. 7. Spotting – Animal's Adaptations, hazardous effects of pollution / Toxicology, Behavioural models, Prepared slides/ models (cell biology, genetics, neurobiology, biotechnology). Life cycles. 8. Comment upon the biological tools a. Autoclave b. Centrifuge c. Colorimeter 9. Assignment / project 10. Class Record 11. Viva-voce	60
Suggested Readings: • P. S. Verma: Practical Invertebrate Zoology • S. S. Lal : Practical Invertebrate Zoology		



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COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	2	1	3	2	2	3	2	2
CO2	3	2	3	3	2	2	2	3	2	2	3	2	2
CO3	3	2	3	3	3	2	2	3	3	3	3	3	2
CO4	3	2	3	3	3	2	2	3	3	3	3	3	3
CO5	3	2	3	3	3	3	2	3	2	3	3	3	3
CO6	2	3	3	3	3	3	3	3	3	3	3	3	3

(3: High Correlation, 2: Moderate Correlation, 1: Low Correlation, 0: No Correlation)